TASK SPECIFICATION: P1-06 PHASE P1 — FOUNDATIONS & DEVOPS INFRASTRUCTURE

Continuous Integration Pipeline (GitHub Actions CI)

Git Branch: `ci/P1-06-github-actions` · Target Pull Request: PR (ci/P1-06-github-actions)

Core Objective & Technical Scope:
Establish enterprise CI workflow: Automated frontend quality checks, backend Maven verification, Docker builds, and compose smoke testing with PR template and Dependabot.

Priority Level	High	Complexity Rating	`M`
Execution Status	Done ✅	Target PR / Commit	`PR (ci/P1-06-github-actions)`
Upstream Dependencies	P1-05		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Author frontend CI job (npm ci, lint, typecheck, vitest, build).
2. Author backend CI job (Java 21 setup, ArchUnit tests, unit tests, JAR packaging).
3. Author Docker verification job validating compose configuration and building test images.
4. Configure standardized GitHub Pull Request template and automated Dependabot scans.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `.github/workflows/ci.yml`
- `.github/pull_request_template.md`
- `.github/dependabot.yml`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- All CI pipeline jobs execute and report green status on PRs.
- Failing unit or architecture tests block merge automatically.

VERIFICATION & TEST COVERAGE STRATEGY

- GitHub Actions automated CI runner verification